

Synoptic CB: Soil Fluxes

2022-2025 Samples

2026-03-31

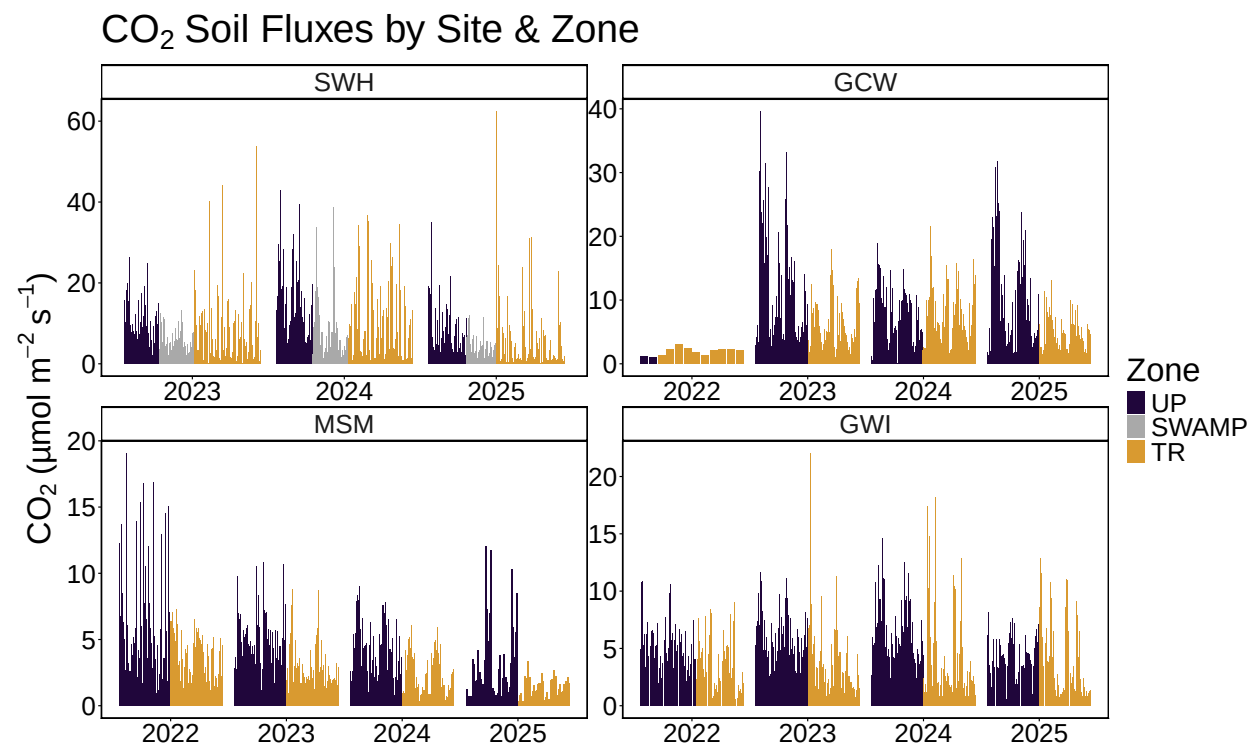
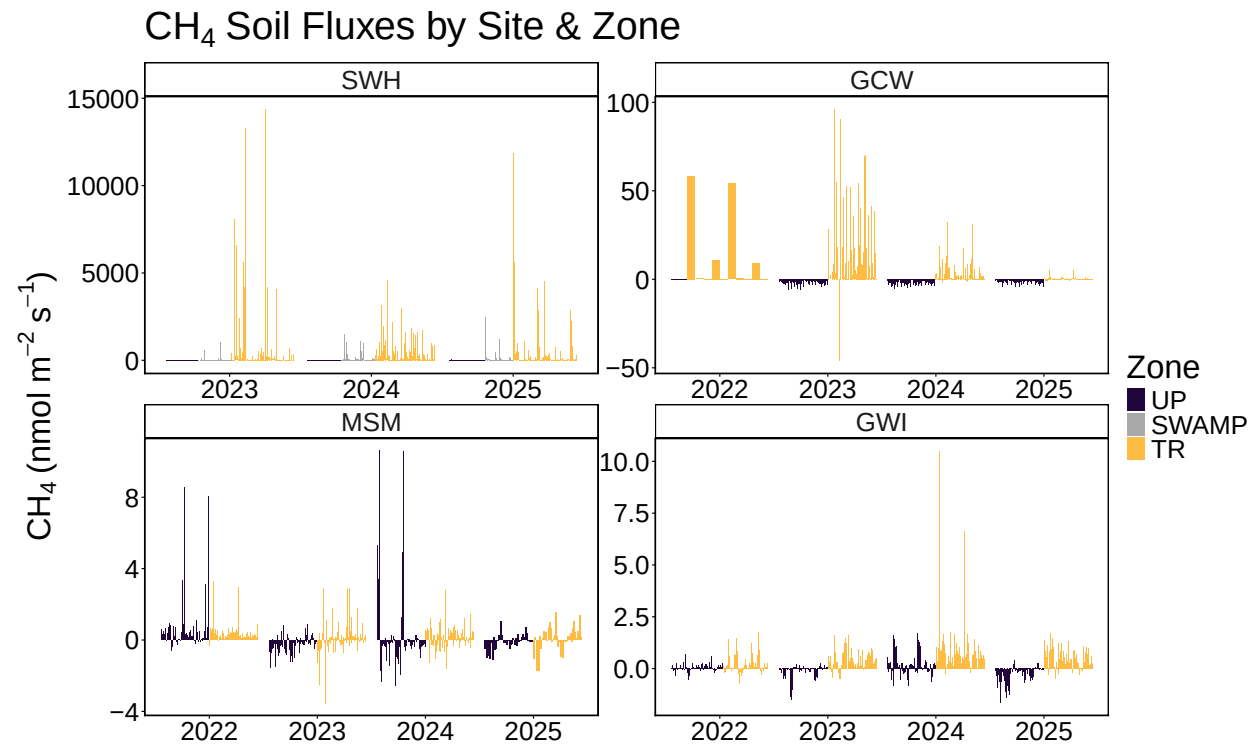
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##Grab Each Year's All Data Files

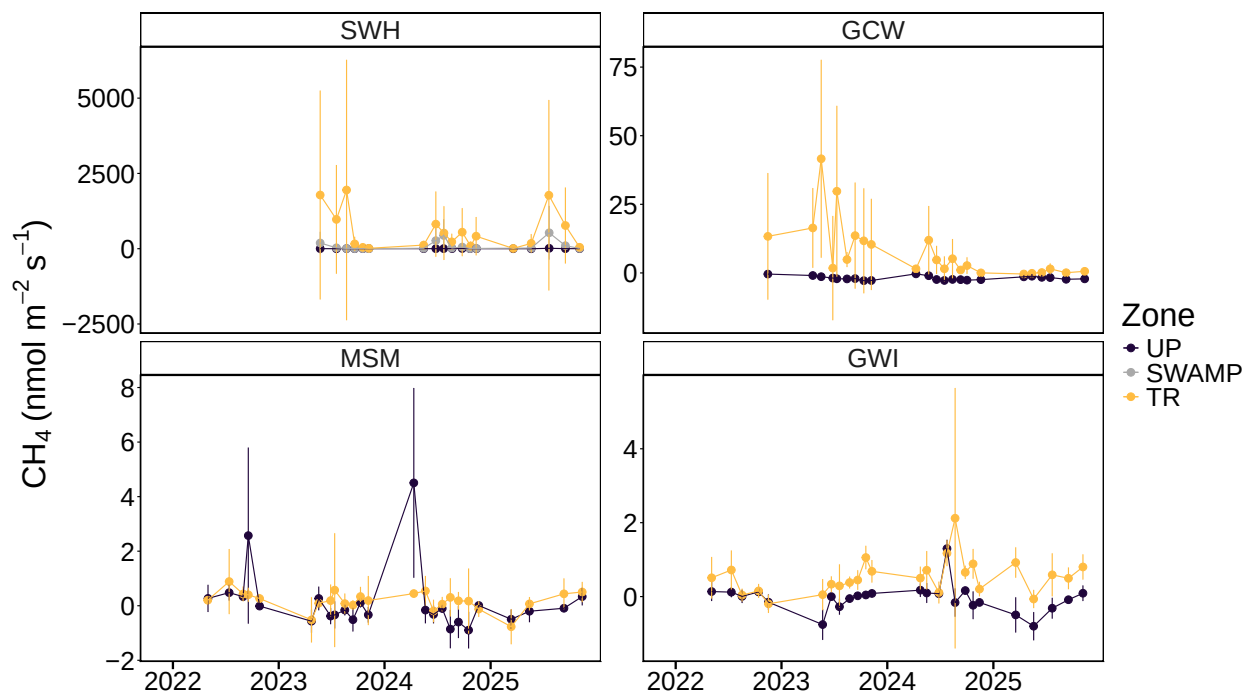
0.1 Write out files

0.2 All Data by Year, Site and Zone

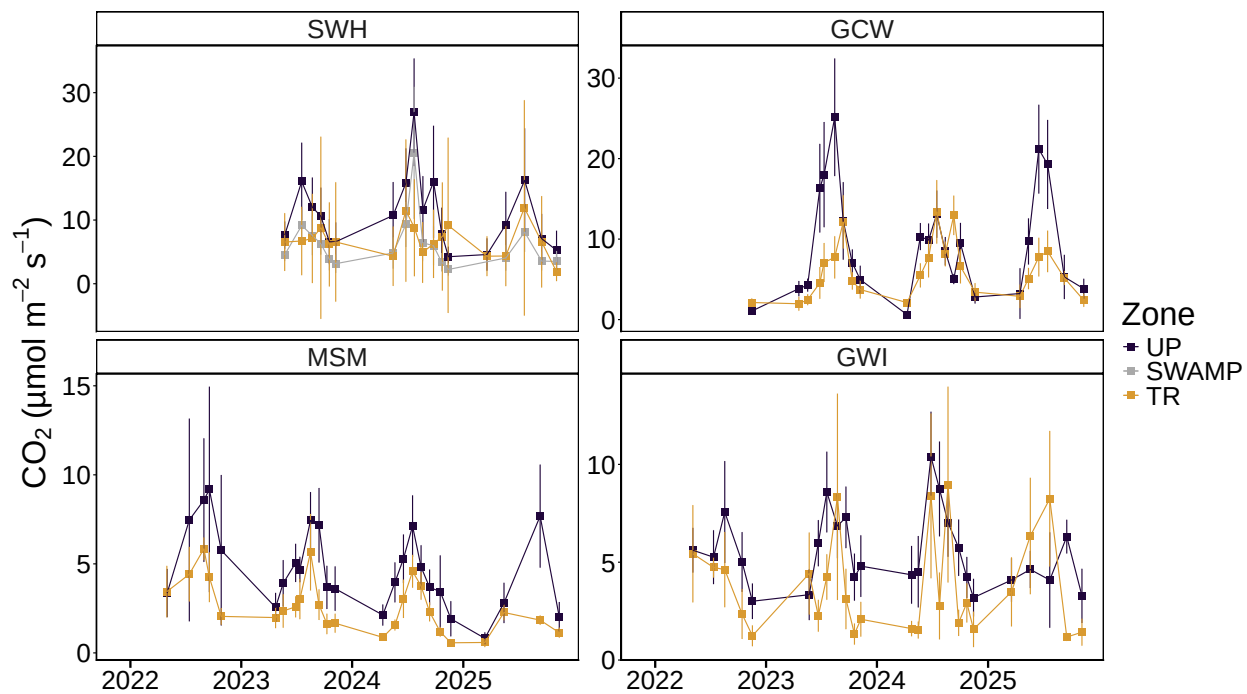


0.3 Summarized data for Site and Zone

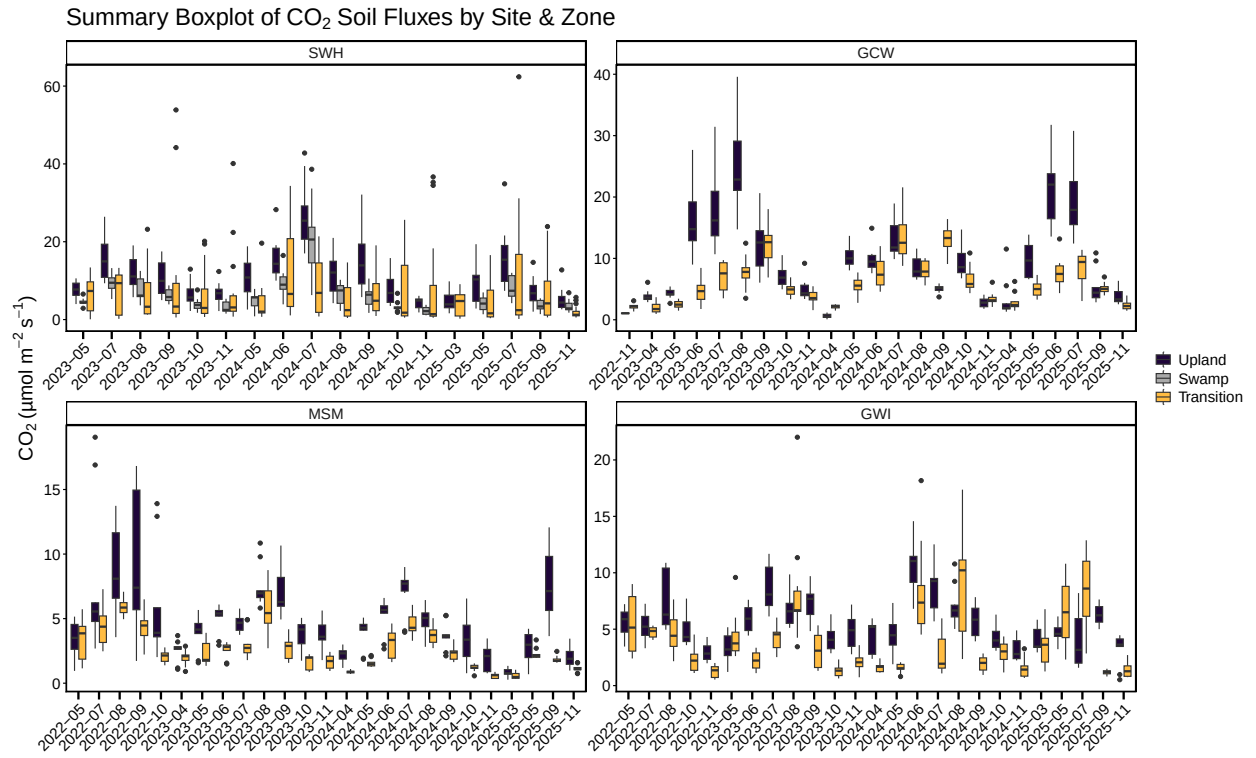
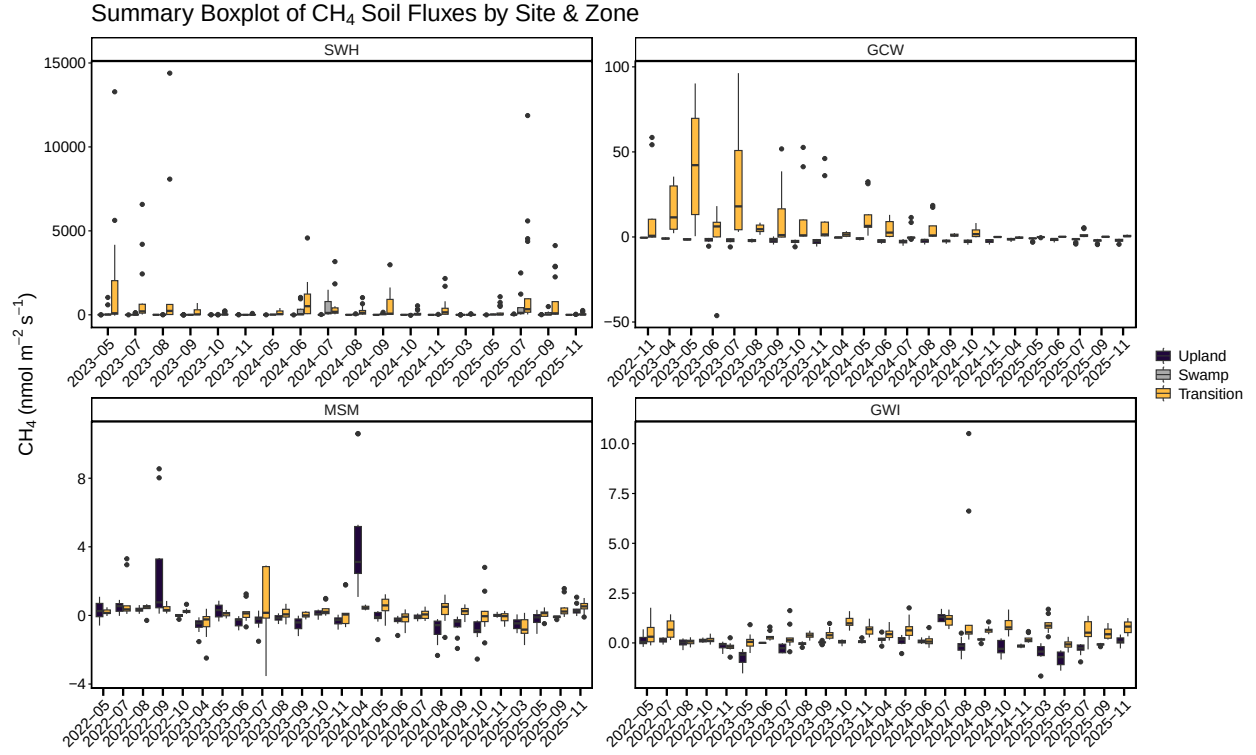
CH₄ Soil Fluxes by Year, Site & Zone



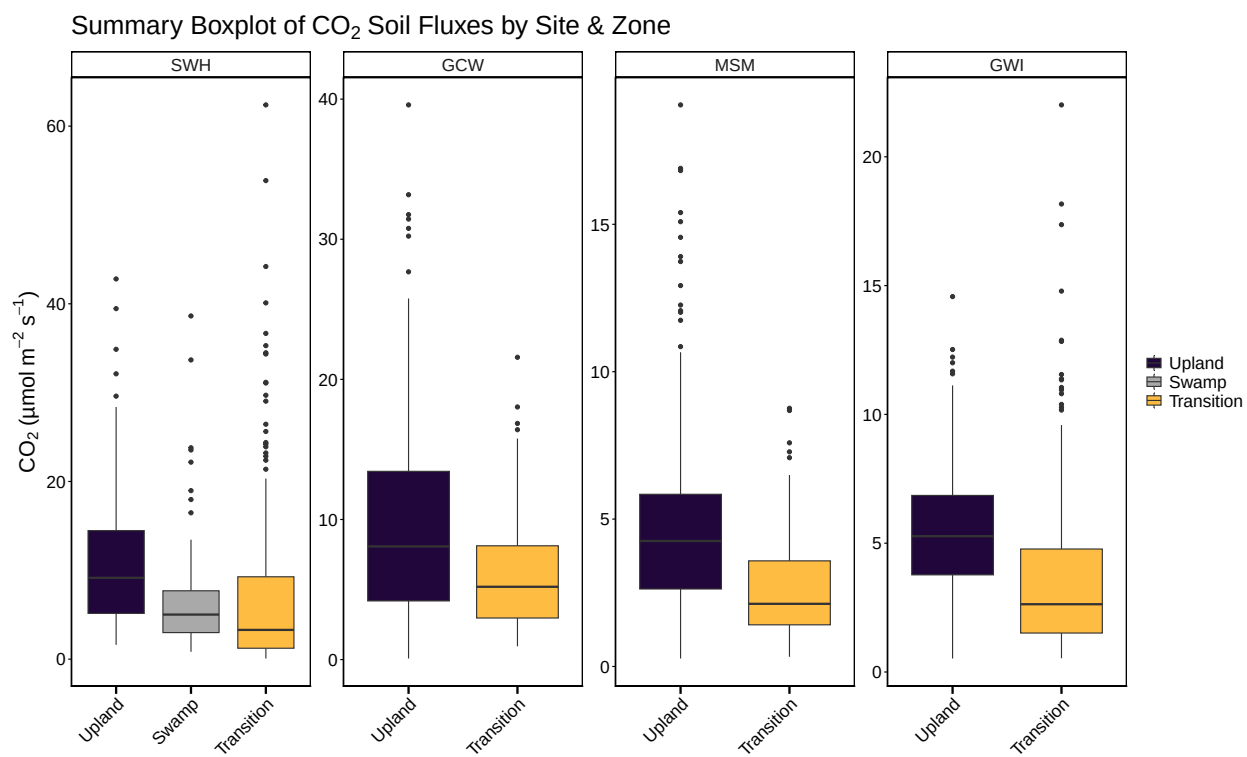
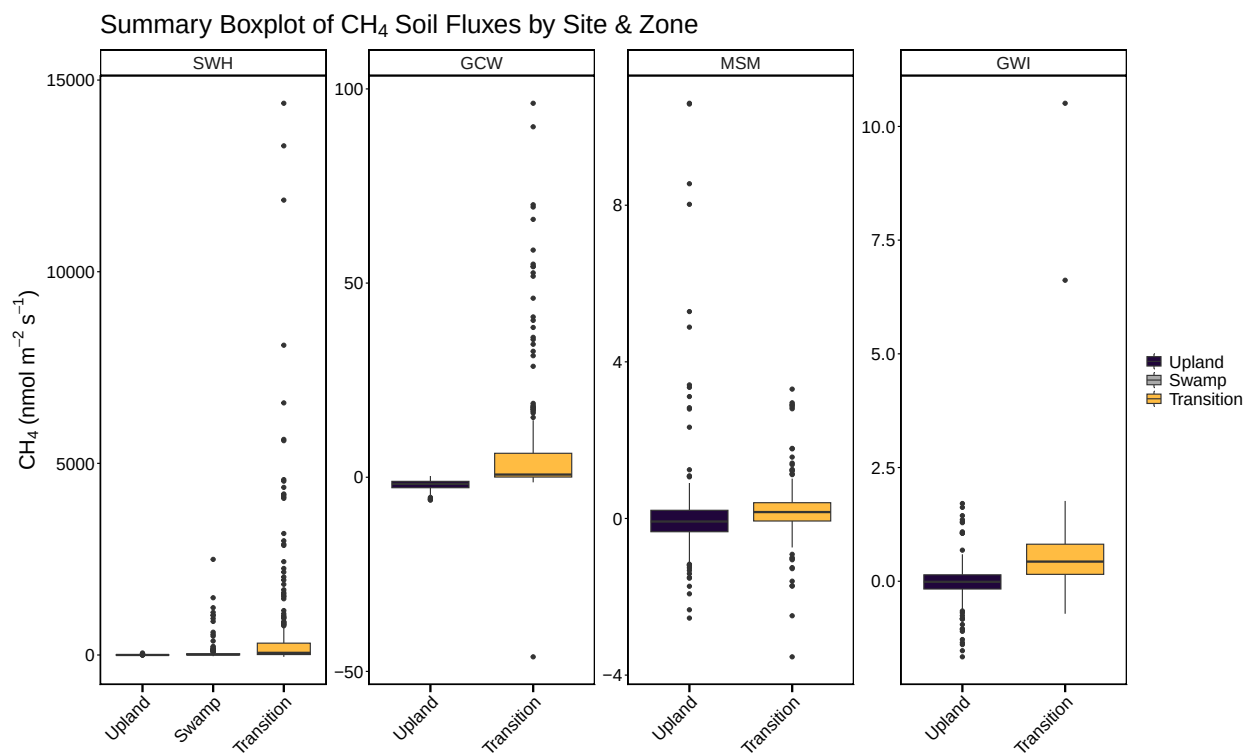
CO₂ Soil Fluxes by Year, Site & Zone



0.4 Boxplot summary by Date



0.5 Boxplot summary by Site



0.6 Calculate summary metrics

0.7 Summary Tables

Table 1: Summary Statistics by Year, Site, and Zone

Year	Site	Zone	Mean CH4 Flux	Min CH4 Flux	Max CH4 Flux	Mean CO2 Flux	Min CO2 Flux	Max CO2 Flux
2022	GCW	Upland	-0.419	-0.544	-0.295	1.060	0.950	1.171
2022	GCW	Transition	13.323	-0.486	58.499	2.123	1.350	3.094
2022	MSM	Upland	0.702	-0.592	8.548	6.672	0.952	19.050
2022	MSM	Transition	0.413	-0.291	3.304	3.930	1.150	7.281
2022	GWI	Upland	0.062	-0.554	0.683	5.402	1.982	10.883
2022	GWI	Transition	0.250	-0.715	1.766	3.852	0.533	9.001
2023	SWH	Upland	1.151	-2.144	7.167	9.914	2.186	26.420
2023	SWH	Swamp	39.684	-7.797	1035.375	5.772	1.528	13.227
2023	SWH	Transition	749.239	-48.271	14394.778	7.007	0.072	53.867
2023	GCW	Upland	-2.017	-5.882	0.317	11.465	2.672	39.584
2023	GCW	Transition	16.427	-46.250	96.303	5.562	0.952	18.030
2023	MSM	Upland	-0.237	-1.525	0.857	4.777	1.080	10.851
2023	MSM	Transition	0.116	-3.534	2.909	2.697	0.861	8.759
2023	GWI	Upland	-0.134	-1.528	0.257	5.868	1.210	11.679
2023	GWI	Transition	0.455	-0.506	1.621	3.686	0.635	22.014
2024	SWH	Upland	2.630	-6.841	30.553	13.315	1.838	42.796
2024	SWH	Swamp	118.074	-30.879	1493.543	7.590	0.830	38.630
2024	SWH	Transition	393.463	-0.416	4575.960	7.444	0.412	36.666
2024	GCW	Upland	-2.161	-5.181	0.086	7.919	0.080	18.945
2024	GCW	Transition	3.366	-1.313	32.435	7.525	1.499	21.575
2024	MSM	Upland	0.198	-2.544	10.603	4.046	0.770	8.996
2024	MSM	Transition	0.174	-1.601	2.801	2.235	0.328	6.065
2024	GWI	Upland	0.157	-0.834	1.709	6.022	1.894	14.573
2024	GWI	Transition	0.795	-0.245	10.509	3.706	0.683	18.168
2025	SWH	Upland	3.110	-3.652	52.260	8.490	1.611	34.878
2025	SWH	Swamp	145.777	-0.247	2494.889	4.630	1.365	12.013
2025	SWH	Transition	553.394	0.049	11868.083	5.713	0.155	62.385
2025	GCW	Upland	-1.732	-4.493	-0.482	10.421	1.316	31.757
2025	GCW	Transition	0.295	-1.014	5.409	5.300	1.477	13.166
2025	MSM	Upland	-0.114	-1.075	1.062	3.332	0.270	12.069
2025	MSM	Transition	0.060	-1.728	1.568	1.458	0.334	3.350
2025	GWI	Upland	-0.322	-1.663	0.421	4.479	0.524	8.200
2025	GWI	Transition	0.548	-0.481	1.692	4.133	0.751	12.879

Table 2: Summary Statistics by Site and Zone

Site	Zone	Mean CH4 Flux	Min CH4 Flux	Max CH4 Flux	Mean CO2 Flux	Min CO2 Flux	Max CO2 Flux
SWH	Upland	2.270	-6.841	52.260	10.841	1.611	42.796
SWH	Swamp	98.320	-30.879	2494.889	6.266	0.830	38.630
SWH	Transition	553.854	-48.271	14394.778	6.818	0.072	62.385
GCW	Upland	-1.973	-5.882	0.317	9.840	0.080	39.584
GCW	Transition	7.172	-46.250	96.303	6.014	0.952	21.575
MSM	Upland	0.118	-2.544	10.603	4.714	0.270	19.050
MSM	Transition	0.189	-3.534	3.304	2.618	0.328	8.759
GWI	Upland	-0.038	-1.663	1.709	5.545	0.524	14.573
GWI	Transition	0.548	-0.715	10.509	3.814	0.533	22.014

0.8 Make Tables into PNGs for README

0.9 Write out summarized data